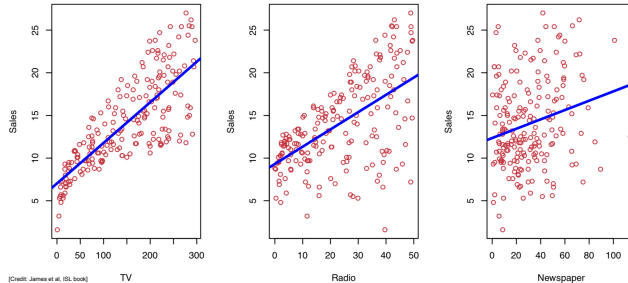


Section 2A. Intro to Statistical Learning

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What is Statistical Learning?

- ▶ **Example:** Consider data collected from 200 marketing campaigns
 - ▶ For each campaign, we know amounts invested in **radio**, **TV**, and **newspaper** advertisement
 - ▶ For each campaign, we also know the total number of **sales**
 - ▶ Using this data, can we predict **Sales** (our output) of a campaign from the **invested amounts** (our inputs)? $\text{Sales} \approx f(\text{Radio, TV, Newspaper})$



Notation

- ▶ We generically denote the input vector by $X = (X_1, \dots, X_p)^\top$ and a particular value of the input vector by $\mathbf{x} = (x_1, \dots, x_p)^\top$. In the previous example, the *input* vector has three entries ($p = 3$):

$$X = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} = \begin{pmatrix} \mathbf{TV} \\ \mathbf{Radio} \\ \mathbf{Newspaper} \end{pmatrix}.$$

Particular values of the input vector X will be denoted by a bold font letter $\mathbf{x} = (x_1, \dots, x_p)^\top \in \mathbb{R}^p$. For example, define $\mathbf{x} = (1000, 800, 900)^\top$; hence, $X = \mathbf{x}$ indicates that the investments are **TV** = 1000, **Radio** = 800, and **Newspaper** = 900.

- ▶ We generically denote the output variable by Y and a particular value of the output by y . For example, define $Y = \mathbf{Sales}$ and $y = 2500$; hence, $Y = y$ indicates that **Sales** = 2500.

Statistical Learning problem

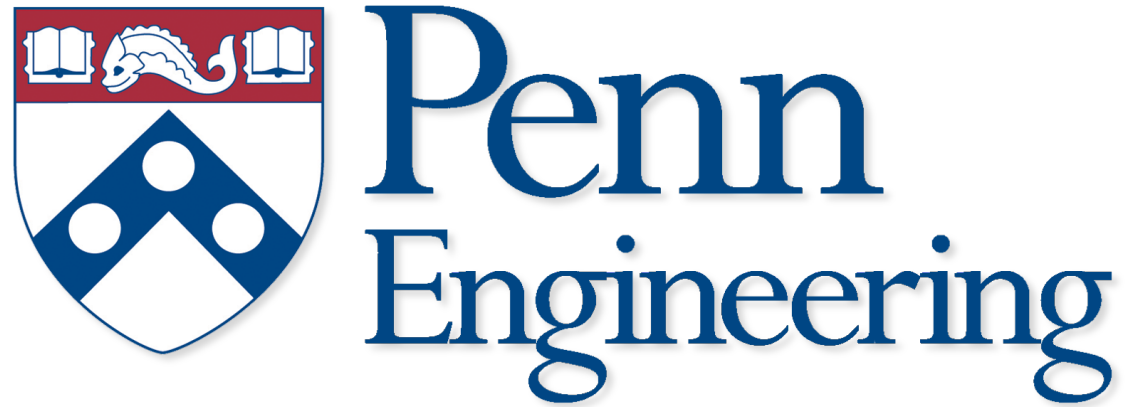
Theoretical setup:

- ▶ In this course, we assume that X is a vector of r.v.'s and Y is a scalar r.v.
- ▶ In the field of statistical learning, we commonly assume that our outputs are generated by an *additive model* of the form

$$Y = f(X) + \varepsilon$$

where f is an *unknown* function called the *regression function* and ε represents the *measurement noise* (which we assume to be a r.v. with zero mean and known variance, $\mathbb{E}[\varepsilon] = 0$ and $\text{Var}[\varepsilon] = \sigma^2$).

Problem: The main problem in statistical learning is to estimate f (the unknown regression function) from random samples drawn from the additive model.



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